Topic 22 — Renewable Energy (नवीकरणीय ऊर्जा)

Covers syllabus (click to expand)

Solar Energy | Wind Energy | Biomass Energy (बायोमास ऊर्जा) | Geothermal Energy (भू-तापीय ऊर्जा) | Tidal Energy | Small Hydro Projects | Green Hydrogen

Sources baked in: NCERT Geo 11–12; MNRE; NAPCC; National Solar Mission; Green Hydrogen Mission 2023; ISA; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — biogas composition; ISA traps; Himalayan hydro; green vs grey hydrogen **Last verified:** August 2026

Current Affairs (this topic)

- India's renewable capacity and green-hydrogen headlines refresh energy CA.
- Keep renewable vs clean vs conventional classification traps stable.

22.1 Solar Energy

Identity: PV + CSP | National Solar Mission 2010 | ISA Gurugram 2015 | 500 GW non-fossil 2030

- **Solar energy** is radiant light and heat from the Sun — harnessed via **photovoltaic (PV)** panels or **concentrated solar power (CSP)** (संकेंद्रित सौर ताप).
- **PV cells** (silicon semiconductors) convert photons to DC electricity; an inverter converts to AC for the grid.
- **CSP** uses mirrors to heat fluid, produce steam, and drive a turbine — can store heat for after-sunset generation.
- Highest insolation (सूर्यताप) in **Rajasthan, Gujarat, Madhya Pradesh, Maharashtra**. **Bhadla Solar Park (Rajasthan)** is among the world's largest.
- **National Solar Mission (2010)** is a pillar of **NAPCC (2008)** — also called **JNNSM** from January 2010 — target scaled to **100 GW solar**, achieved around **2022–2024**.
- Solar has the **greatest renewable potential** in India among common options; solar cells are the most eco-friendly among petroleum / forest products / nuclear fission / solar options.
- **500 GW non-fossil fuel capacity by 2030** (COP26, 2021) includes solar, wind, hydro, nuclear, and biomass — solar remains the largest growth component.
- **International Solar Alliance (ISA)** (अंतरराष्ट्रीय सौर गठबंधन) launched **30 November 2015** at **COP21 Paris** by **India and France** — treaty-based IGO, HQ **Gurugram, Haryana**.
- ISA membership is **not** "all UN members" — historically focused on countries between the Tropics of Cancer and Capricorn (membership has since widened). First Summit: **New Delhi, 11 March 2018**. Declaration ambition: mobilise about **USD 1 trillion** for solar by **2030**.

- **Domestic Content Requirement (DCR)** in solar RfS means using domestically manufactured PV cells and modules.
- Silicon wafers / cells are largely imported or assembled — India is **not** among the top silicon-wafer manufacturers; solar tariffs are regulated by **CERC/SERCs**, not set solely by SECI.
- **OSOWOG (One Sun One World One Grid)** and **World Solar Bank** concepts are linked to ISA's cross-border solar grid vision.
- **PM-KUSUM (2019):** solar pumps for farmers and grid-connected solar on barren/fallow land — MNRE nodal.
- **Cochin International Airport** was the **first airport in the world** fully powered by solar energy.
- **Diu** became the first Indian UT / smart-city example to run on **100% solar** (daytime / UT stems). **Surat** became the first district with **100% solar-powered Primary Health Centres**.
- **Bhadla (Rajasthan)** is India's mega solar park fact — not "Gujarat has the largest solar park" in newer stems. India's large floating solar example: **Ramagundam (Telangana)**.

PYQ — UPPCS Prelims 2025, Q109

ISA: (1) First treaty-based IGO in India with 38 member countries. (2) World Solar Bank and OSOWOG under ISA.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: A — OSOWOG/World Solar Bank linked to ISA is correct. **"38 members" is outdated** — signatories far exceed 38.

Scheme	Year	Fact
National Solar Mission	2010	NAPCC pillar / JNNSM
ISA	2015	HQ Gurugram; Summit 2018 Delhi
PM-KUSUM	2019	Solar pumps
PLI Solar PV	2021	Domestic manufacturing

> **Logic:** ISA HQ = **Gurugram** — trap: New Delhi or Paris. DCR = domestic solar manufacturing.

22.2 Wind Energy

Identity: Kinetic air → turbines | Power ∝ v^3 | Tamil Nadu leader | Onshore dominant

- **Wind energy** converts the **kinetic energy of moving air** into electricity using **wind turbines**.
- Power output depends on **wind speed cubed (v^3)** — doubling wind speed gives roughly **8× power**.
- Utility-scale sites need consistent annual average wind **>6 m/s** — coastal and open plateau areas preferred.
- **Tamil Nadu** is historically India's leading wind state (**Muppandal wind farm**). **Gujarat, Karnataka, Maharashtra, Rajasthan** follow.
- **Onshore** turbines dominate India. **Offshore wind** potential on **Gujarat and Tamil Nadu coasts** — stronger winds but higher cost.
- **Intermittency:** wind is variable — needs **grid integration**, **hybrid wind+solar**, or **storage** (batteries, pumped hydro).
- **Capacity factor:** wind farms typically produce **~25–35%** of nameplate capacity annually — location-dependent.
- **UP has limited wind potential** compared to western and coastal states — focus is national leaders.

Feature	Onshore	Offshore
Cost	Lower	Higher
Wind quality	Moderate (नरम दल)	Stronger, steadier
India status	Mature	Early policy stage

> **Logic:** Rajasthan leads **solar**, not wind. **Tamil Nadu** = wind leader trap.

22.3 Biomass Energy

Identity: Organic matter → energy | Biogas = CH_4 + CO_2 | Bagasse UP sugar belt | Anaerobic digestion

- **Biomass energy** is stored **chemical energy in organic matter** — plants, crop residue, animal waste, wood — released by burning, fermentation, or **anaerobic digestion**.
- **Anaerobic digestion:** bacteria break organic matter **without oxygen** in a digester → **biogas** + **slurry** (organic fertiliser).
- **Biogas composition:Methane (CH_4) ~50–70%** + **Carbon dioxide (CO_2) ~25–45%** + traces of H_2S and moisture.
- **Butane** is an LPG component — **not** a major biogas constituent. **Carbon monoxide** is incomplete combustion product — not main biogas component.
- **Bagasse** (sugarcane crush residue) is burned in **UP sugar mills** (Meerut (मेरठ), Muzaffarnagar belt) for **cogeneration** — electricity + process heat.
- **GOBAR-Dhan / SATAT:** government push for community biogas and **bio-CNG** — oil marketing companies procure compressed biogas.
- **E20 ethanol blending target by 2025** — ethanol from sugarcane, maize, and agri-waste reduces petrol imports.
- **Stubble burning** wastes biomass and causes **PM2.5 pollution** — not sustainable energy use.

PYQ — UPPCS Prelims 2025, Q28

Major constituents of biogas: (1) Methane (2) Butane (3) Carbon dioxide (4) Carbon monoxide

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

Show answer

Ans: A — **Methane + Carbon dioxide**. Butane is the trap.

> **Logic:** Biogas = **CH_4 + CO_2** — not butane.

22.4 Geothermal Energy

Identity: Earth's interior heat | Puga Ladakh (लद्दाख) | Baseload potential | NOT from Sun

- **Geothermal energy** harnesses **heat from Earth's interior** — hot rocks, steam, or underground water — for electricity or direct heating.
- Heat sources: **radioactive decay** of minerals + residual heat from Earth's formation — temperature rises **~25–30°C per km** depth.
- Wells tap hot reservoirs → steam/hot water spins **turbine** — or **direct use** for heating (greenhouses, spas).
- **India's prospects / provinces:** Himalayas, Cambay, West Coast, **SONATA (Son–Narmada–Tapi)**, Godavari, Mahanadi, Sohana — **Ganga Delta is not** a listed geothermal province.
- Key sites: **Puga Valley (Ladakh)** — highest potential; **Manikaran (HP)**; **Tattapani (Chhattisgarh)**; **Cambay Basin (Gujarat)**.
- **No large-scale commercial geothermal power plant** operational in India yet — pilot exploration only (GSI/MNRE).
- **Advantages:baseload renewable** — can run 24/7 unlike solar/wind at a suitable site.
- **Global leaders:** Iceland, USA, Indonesia, Philippines — India remains early stage.
- **NOT derived from the Sun** in the "which is not from the Sun" MCQ sense — unlike biomass, wind, or solar.

> **Logic:** Geothermal = **Earth heat**, not sunlight. Iceland benchmark ≠ India primary source.

22.5 Tidal Energy

Identity: Moon's gravity | Khambhat highest range | Predictable | No major Indian plant yet

- **Tidal energy** harnesses the **rise and fall of sea levels** caused primarily by **gravitational pull of the Moon** (and Sun).
- **Tidal barrage:** dam (दाम -) across estuary (ज्वारनदमुख) — gates allow water in at high tide, release through **turbines** at low tide.
- **Tidal stream turbines:** underwater turbines in fast tidal channels — no large dam required.
- **India's potential:Gulf of Khambhat (Cambay)** — tidal range up to **~11 m**, highest in India. **Gulf of Kutch** is second.
- **No major commercial tidal power plant** operational in India — papers test **potential**, not installed leadership.
- **Advantages:highly predictable** (lunar calendar) — unlike variable wind/solar.
- **Disadvantages:** high capital cost, estuary ecosystem (पारिस्थितिकी तंत्र) disruption, siltation in Khambhat.
- **Tidal ≠ wave energy:** tidal = gravity-driven sea level change; wave = wind-driven surface waves.

PYQ — UPPCS Prelims 2022

Largest tidal energy producing area in India:

A. North Circars

B. Mannar

C. Khambhat Coast

D. Kerala

Show answer

Ans: C — **Gulf of Khambhat (Cambay)** has the highest tidal range.

> **Logic:** Tidal potential = **Gujarat coast** (Khambhat/Kutch), not Thar (थार) Desert.

22.6 Small Hydro Projects

Identity: ≤25 MW MNRE | Himalayan potential | Ramganga 198 MW Garhwal | Run-of-river vs storage

- **Hydropower** converts **potential/kinetic energy of falling water** into electricity — renewable, rainfall-dependent.
- **MNRE definition:Small hydro project (SHP) = ≤25 MW**. Above 25 MW = **large hydro**.
- **Run-of-river:** minimal large reservoir — lower submergence, preferred in sensitive Himalaya.
- **Storage/pondage:** dam + reservoir (Tehri (टिहरी), Bhakra model) — higher head but displacement/seismic risks.
- **Himalayan rivers** have **enormous hydroelectric potential** — steep gradient, snow-fed perennial flow.
- **Ramganga Multipurpose Project:Garhwal (Uttarakhand)**, dam on Ramganga tributary of Ganga, **~198 MW** hydropower — irrigates western **UP** plains.

- **Pumped storage hydro:** uses off-peak electricity to pump water uphill, releases at peak demand — grid balancing with solar/wind.

PYQ — UPPCS Prelims 2025, Q31

Hydroelectric power in Northern India: (1) Himalayan region has enormous hydroelectric potential. (2) Eastern UP and Bihar cannot utilize this because Himalayas to their north lie in Nepal.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: D — Stmt 1 true. Stmt 2 false — major Indian hydro in **Uttarakhand, HP, Sikkim, Arunachal**; UP benefits via Ganga/Yamuna (यमुना) systems.

PYQ — UPPCS Prelims 2024, Q72

Ramganga Project — which pair is NOT correctly matched?

Options include: Dam length 825.8 m | Hydropower **198 MW** | Location Garhwal | Irrigated area 6 lakh ha

Show answer

Verify all four (चातुर्याम) stats — **198 MW** and **Garhwal** (गढ़वाल) are standard correct facts. Irrigated area figure is the common mismatch trap in options.

Category	Capacity
Small hydro (SHP)	≤25 MW
Large hydro	>25 MW
Ramganga	~198 MW (large)

> **Logic:** SHP cap = **25 MW**. Singrauli = **thermal**, not hydro.

22.7 Green Hydrogen

Identity: Electrolysis + renewable power | Mission 2023 | 5 MMT/yr 2030 | Green vs grey vs blue

- **Hydrogen (H_2)** is an **energy carrier** — must be produced from water, gas, or biomass. Burns to water vapour with no carbon at point of use.
- **Green hydrogen:electrolyser** splits **H_2O → H_2 + O_2** powered by **renewable electricity** (solar/wind/hydro) — near-zero lifecycle CO_2.

- **Grey hydrogen:** ~98% of current global production — **natural gas steam reforming** — releases large **CO_2. NOT green.**
- **Blue hydrogen:** grey production + **carbon capture and storage (CCS)** — lower emissions, still fossil-dependent.
- **Black/brown hydrogen:** from **coal gasification** — very high emissions.
- **National Green Hydrogen Mission (January 2023):** target **5 MMT green hydrogen per annum by 2030** — MNRE coordinates.
- **National Hydrogen Energy Board** was set up on **3 October 2003** to promote hydrogen energy (older bank fact; Mission 2023 is the current flagship).
- Green hydrogen can be used **directly in internal combustion**, **blended with natural gas** for heat/power, and in **hydrogen fuel cells** for vehicles — all three statements are correct in recent stems.
- **Fuel cells:** electrochemical conversion of H_2 + O_2 → electricity; by-products **heat and water**; generate **Direct Current (DC)**; usable for buildings **and** portable devices (laptops etc.).
- **Microbial fuel cells (MFC):** living microbes as catalysts convert organic substrates to electricity; attractive in **wastewater treatment** (clean water + power). Substrates are mainly organic (glucose, etc.), not "inorganic-only."
- **Sectors:** refineries, **ammonia/fertiliser (urea)**, **green steel**, heavy transport, shipping.
- **Challenges:** high cost vs grey H_2, energy conversion losses, storage/transport infrastructure (compress, liquefy, pipeline).
- **Algae biofuels:** possible on continents as well as seas; developing-country limits = high expertise + large facilities with ecological/social concerns.
- **Bioasphalt:** non-petroleum renewable binder for roads — lowers fossil dependence and can reduce surface temperature versus conventional asphalt.

Type	Source	CO_2
Green	Renewable electrolysis	Very low
Grey	Natural gas	High
Blue	Gas + CCS	Moderate

> **Logic:Green = renewable electrolysis only.** Mission launched **2023**, not 2019. Fuel cell output = **DC**.